

Kinematics of the wind field

(Kinematics: from the Greek word for 'motion', a description of the motion of a particular field without regard to how it came about or how it will evolve)

In this section of the course, our goal will be to

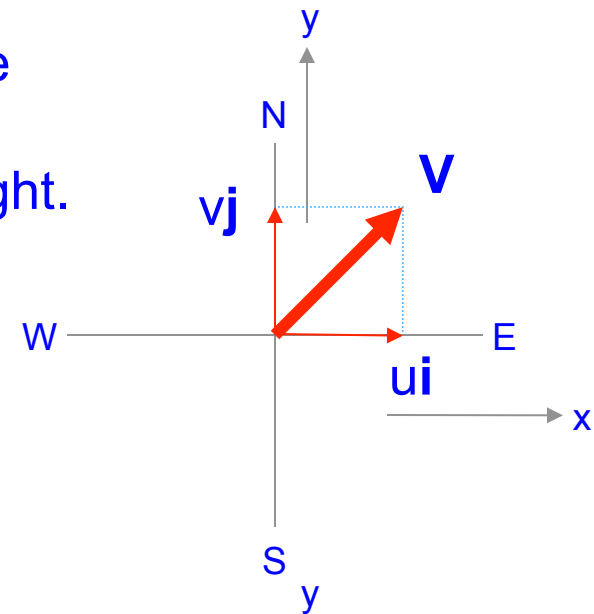
- 1) develop a mathematical and an intuitive understanding of the kinematic properties of the wind field

and

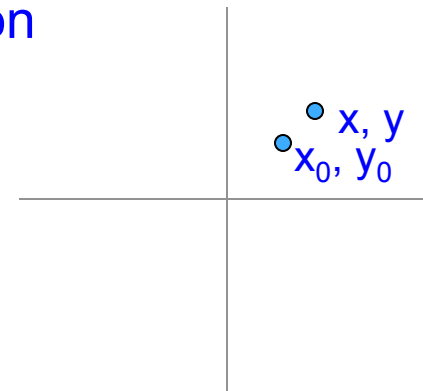
- 2) learn how these kinematic properties can be used to understand the evolution of weather systems

See Bluestein Vol. I, Sections 3, 5.5 for more information

To derive a mathematical expression for the key kinematic properties of the wind field we will use the coordinate system on the right.



We will use a technique called Taylor Expansion to estimate the wind field at an arbitrary point x, y from the wind at a nearby point x_0, y_0



$$u = u_0 + \frac{1}{2} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) x - \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) y + \frac{1}{2} \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right) x + \frac{1}{2} \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) y$$

$$v = v_0 + \frac{1}{2} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) y + \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) x - \frac{1}{2} \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right) y + \frac{1}{2} \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) x$$



Translation

Relative
Vorticity

Stretching
Deformation

Shearing
Deformation

Any wind field that varies linearly can be characterized by these five distinct properties. Non-linear wind fields can be closely characterized by these properties.

$$u = u_0 + \frac{1}{2} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) x - \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) y + \frac{1}{2} \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right) x + \frac{1}{2} \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) y$$

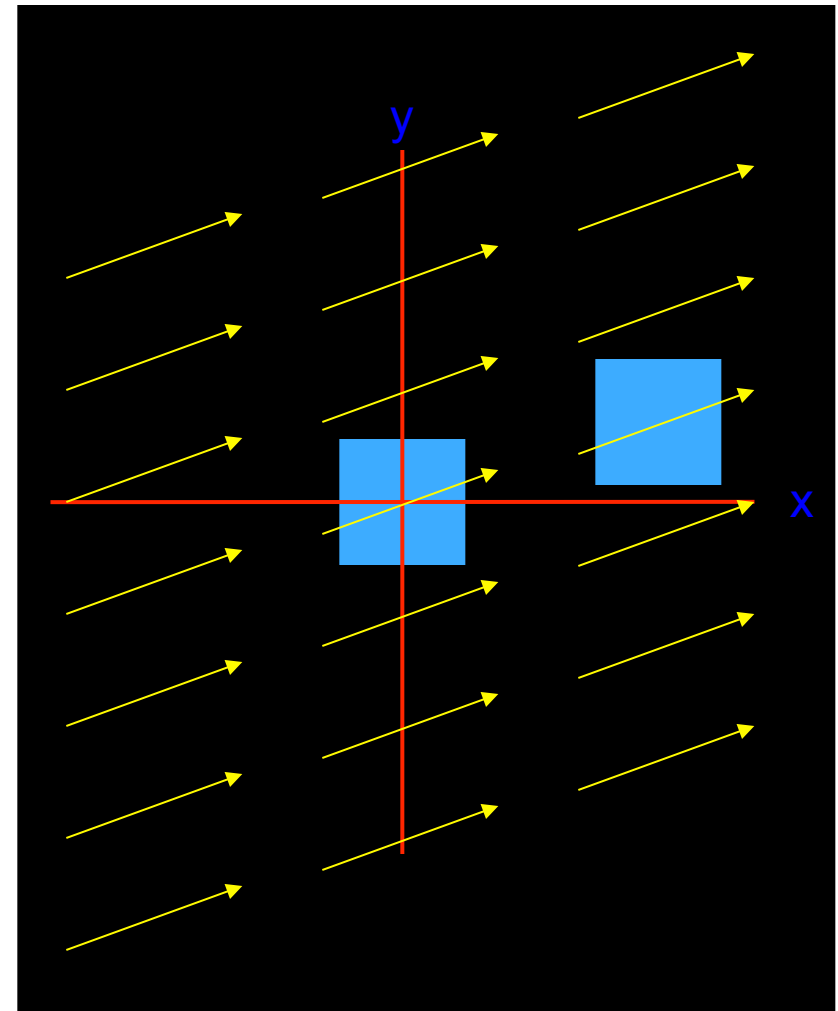
$$v = v_0 + \frac{1}{2} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) y + \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) x - \frac{1}{2} \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right) y + \frac{1}{2} \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) x$$



Translation

The effect of translation on a fluid element:

Change in location, no change in
area, orientation, shape



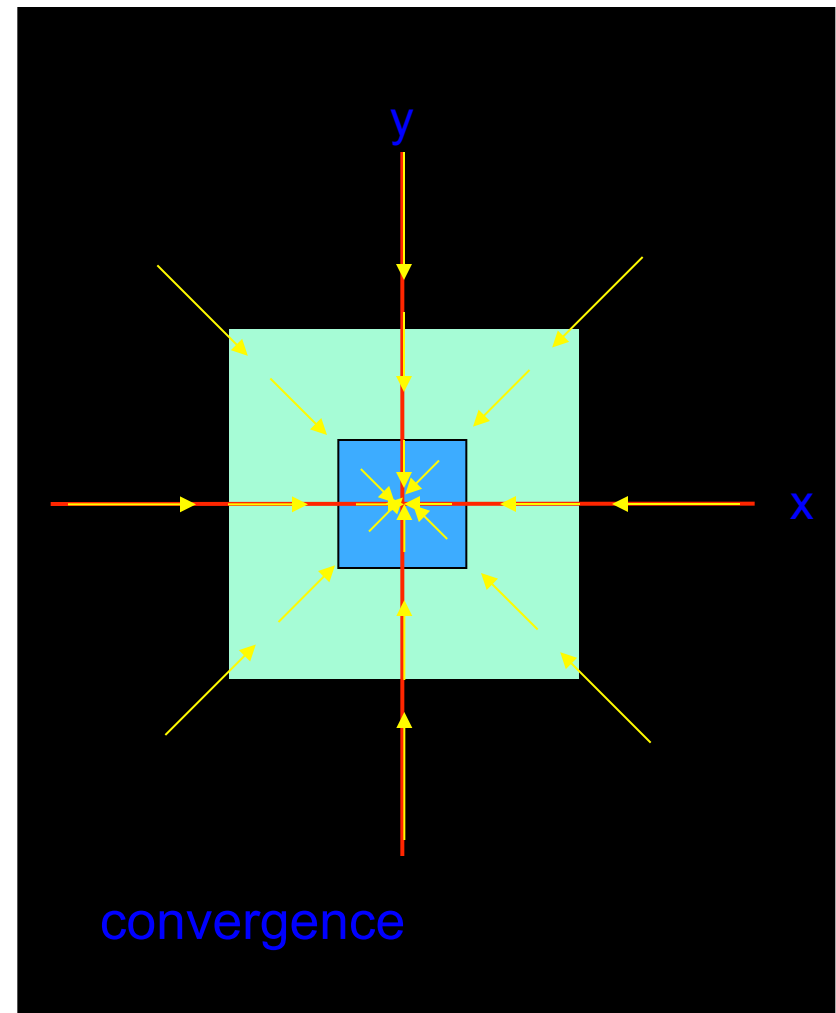
$$u = u_0 + \frac{1}{2} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) x - \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) y + \frac{1}{2} \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right) x + \frac{1}{2} \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) y$$

$$v = v_0 + \frac{1}{2} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) y + \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) x - \frac{1}{2} \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right) y + \frac{1}{2} \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) x$$

↑
Divergence ($\delta > 0$)
Convergence ($\delta < 0$)

The effect of convergence on a fluid element:

Change in area, no change in
orientation, shape, location



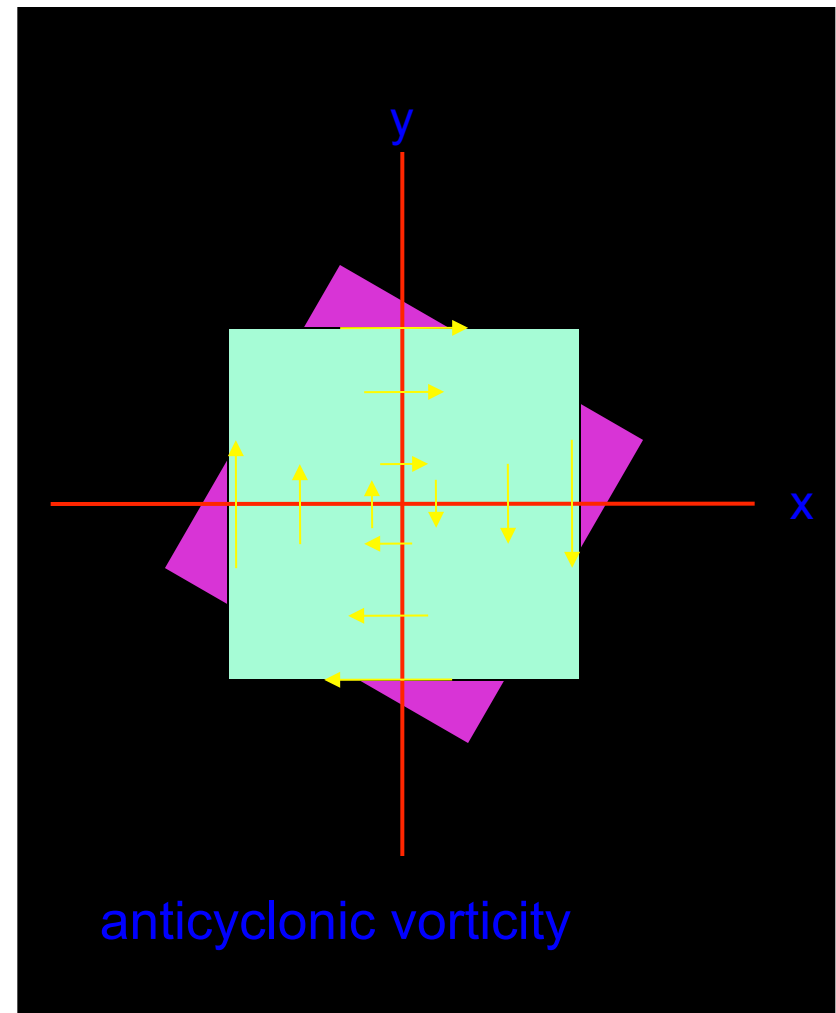
$$u = u_0 + \frac{1}{2} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) x - \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) y + \frac{1}{2} \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right) x + \frac{1}{2} \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) y$$

$$v = v_0 + \frac{1}{2} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) y + \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) x - \frac{1}{2} \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right) y + \frac{1}{2} \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) x$$

Positive (cyclonic) vorticity ($\zeta > 0$).
Negative (anticyclonic) vorticity ($\zeta < 0$)

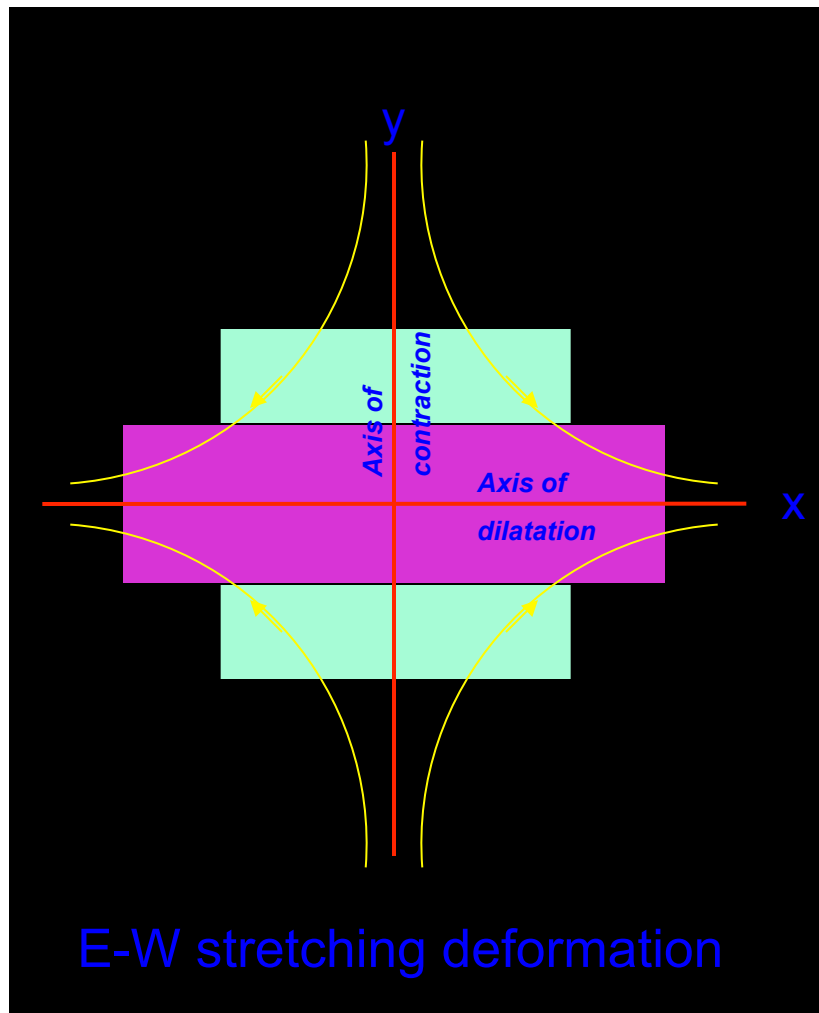
The effect of negative vorticity on a fluid element:

Change in orientation, no change in area, shape, location



$$u = u_0 + \frac{1}{2} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) x - \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) y + \frac{1}{2} \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right) x + \frac{1}{2} \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) y$$

$$v = v_0 + \frac{1}{2} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) y + \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) x - \frac{1}{2} \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right) y + \frac{1}{2} \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) x$$



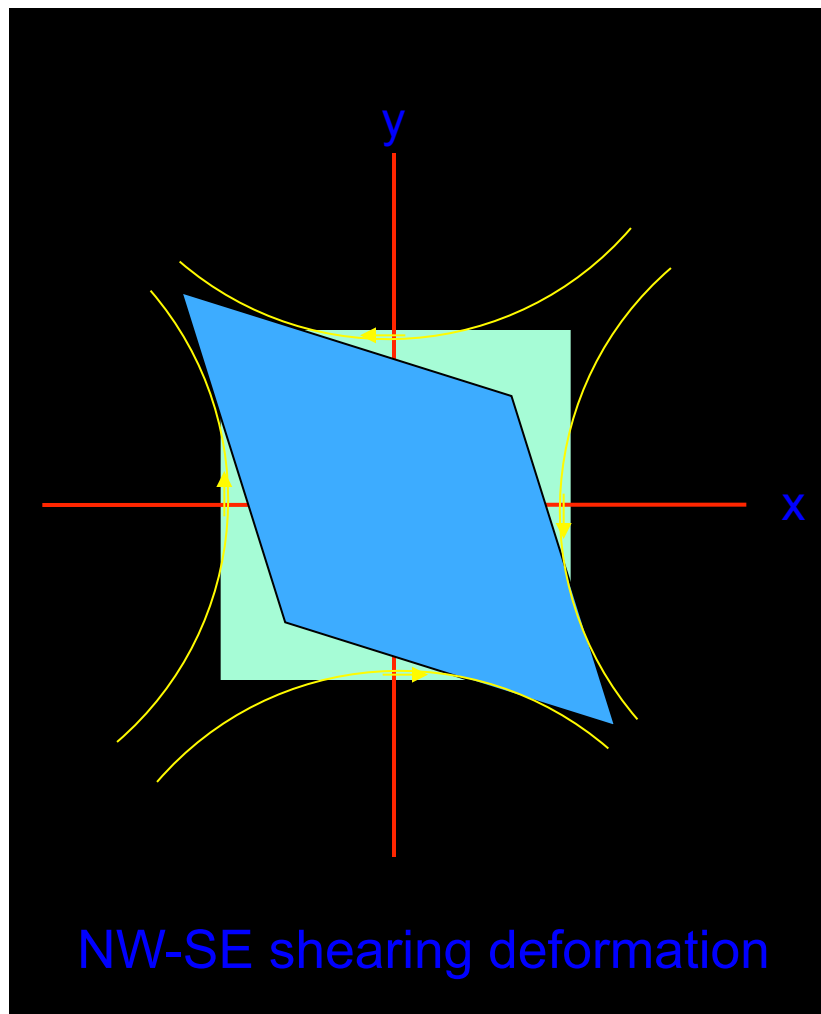
E-W Stretching Deformation ($D_1 > 0$).
N-S Stretching Deformation ($D_1 < 0$).

The effect of stretching deformation on a fluid element:

Change in shape, no change in area, orientation, location

$$u = u_0 + \frac{1}{2} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) x - \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) y + \frac{1}{2} \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right) x + \frac{1}{2} \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) y$$

$$v = v_0 + \frac{1}{2} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) y + \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) x - \frac{1}{2} \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right) y + \frac{1}{2} \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) x$$



SW-NE Shearing Deformation ($D_2 > 0$).
 NW-SE Shearing Deformation ($D_2 < 0$).

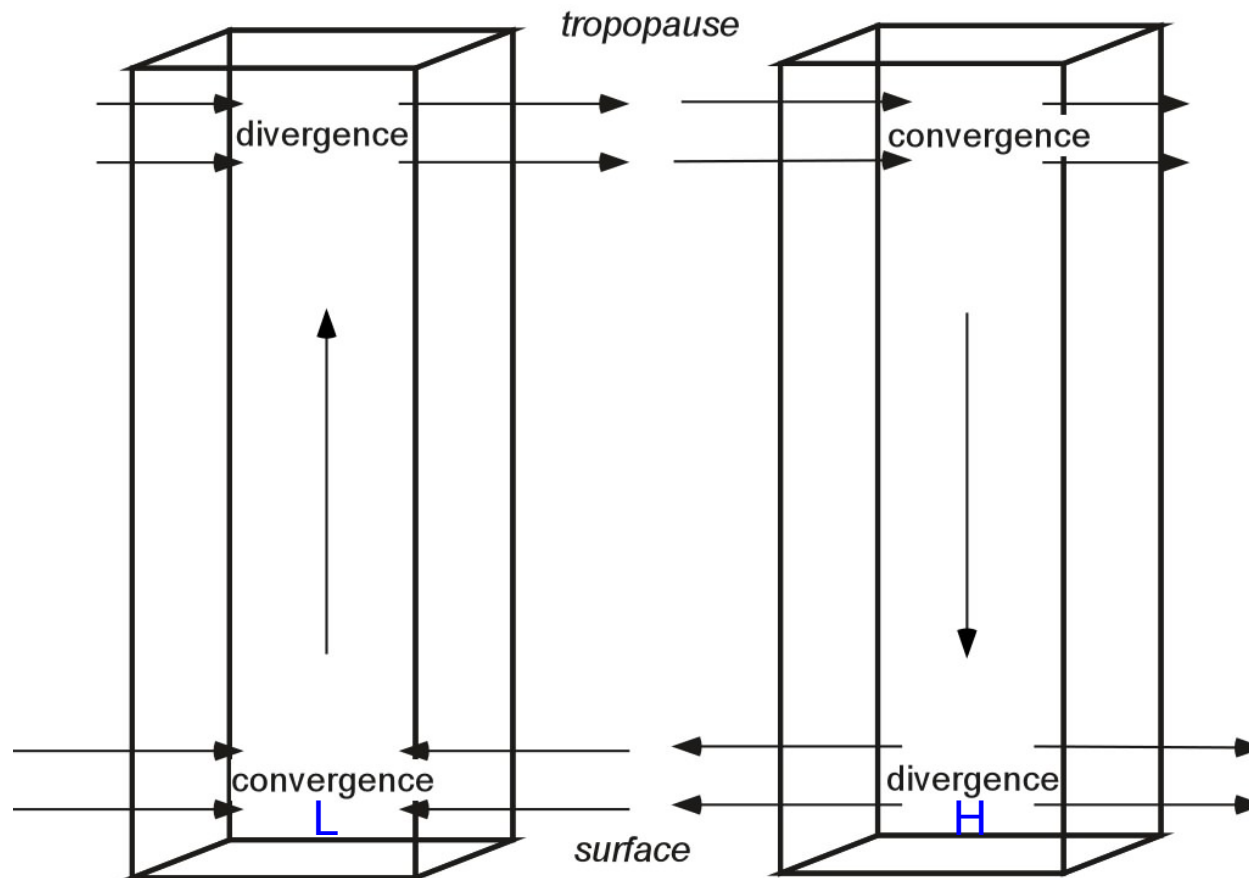
The effect of shearing deformation on a fluid element:

Change in shape, no change in area, orientation, location

Why are we interested in these properties?

We can use and divergence/convergence and the **continuity equation** to diagnose synoptic scale vertical motion within a column.

Vertical motion is what we are interested in in weather forecasting!



The continuity equation and vertical motion

Since synoptic scale vertical motions are difficult to observe, a conceptual model of vertical motions and circulations is needed to properly diagnose synoptic scale vertical motion.

The continuity equation states that

$$\nabla \bullet \mathbf{V} = 0 \quad \text{where} \quad \nabla = \frac{\partial}{\partial x} \hat{\mathbf{i}} + \frac{\partial}{\partial y} \hat{\mathbf{j}} + \frac{\partial}{\partial p} \hat{\mathbf{k}} \quad \&$$

$$\mathbf{V} = u\hat{\mathbf{i}} + v\hat{\mathbf{j}} + \omega\hat{\mathbf{k}}$$

We can also write

$$\nabla_H \bullet \mathbf{V}_H + \frac{\partial \omega}{\partial p} = 0 \quad \text{where} \quad \nabla_H = \frac{\partial}{\partial x} \hat{\mathbf{i}} + \frac{\partial}{\partial y} \hat{\mathbf{j}} \quad \&$$

$$\mathbf{V}_H = u\hat{\mathbf{i}} + v\hat{\mathbf{j}}$$

Note $\nabla \bullet \mathbf{V} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial \omega}{\partial p}$ & $\nabla_H \bullet \mathbf{V}_H = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}$

3-D divergence

Horizontal divergence

From the previous page $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial \omega}{\partial p} = 0$

Let's diagnose the vertical motion (w) by examining the continuity equation. Integrating the above downward from the top of the atmosphere (p_0) to some pressure level p yields

$$\omega(p) - \omega(p_0) = - \int_{p_0}^p \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) \partial p$$

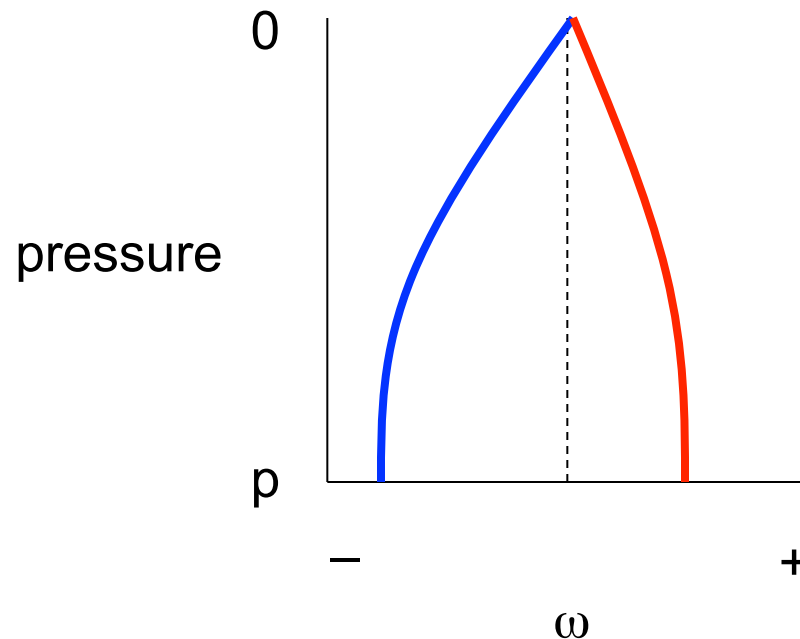
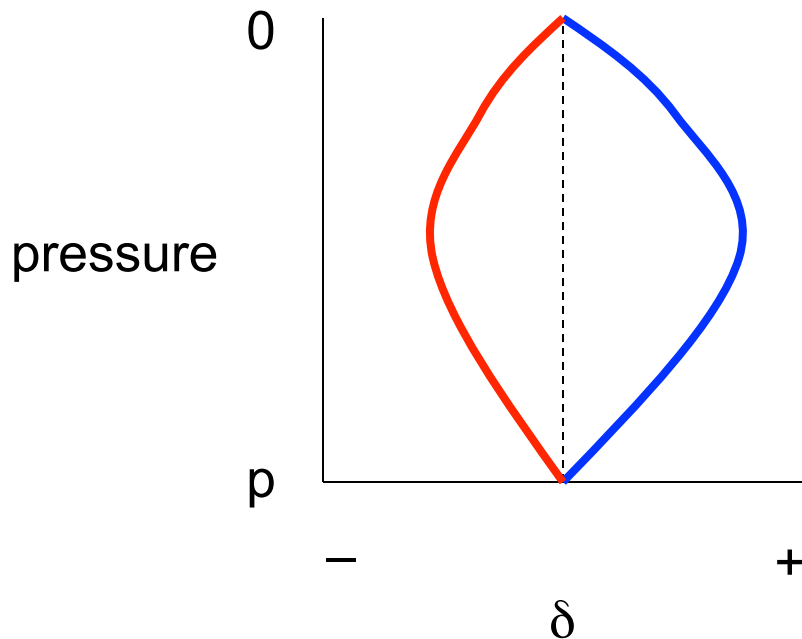
At p_o (top of atmosphere), $p=0$ and $\omega=Dp/Dt=0$ (no vertical motion), so

$$\omega(p) = - \int_{p_0}^p \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) \partial p$$

$$\omega(p) = - \int_{p_0}^p \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) dp$$

What does this equation tell us?

It says that **omega at a given pressure level is proportional to the integral of divergence or convergence above a given pressure level.**



Assuming a mean divergence, we may solve the integral and write

$$\omega(p) = -\overline{\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right)}(p - p_0)$$

Since $p_0=0$, we obtain

$$\omega(p) = -\overline{\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right)}p$$

Interpreting this result:

1) If we have mean divergence aloft,

$$\overline{\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right)} > 0, \text{ since } p > 0, \text{ then } \omega(p) < 0, \text{ thus rising motion.}$$

2) If we have mean convergence aloft,

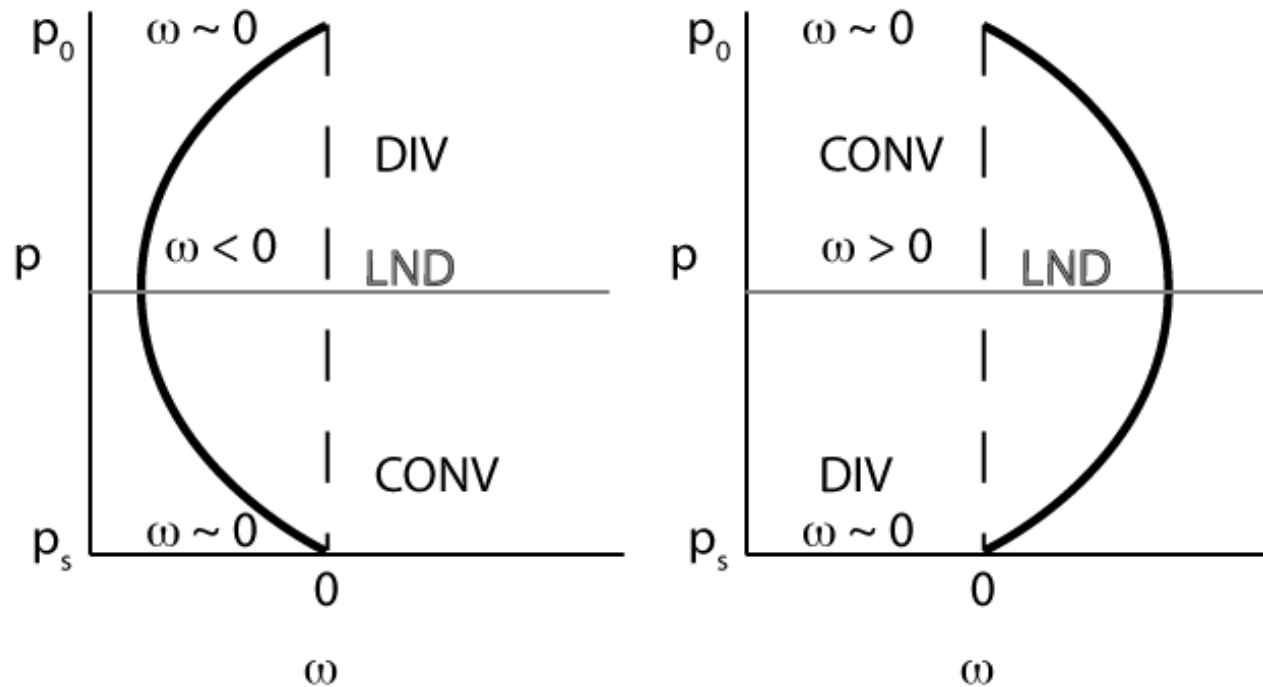
$$\overline{\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right)} < 0, \text{ since } p > 0, \text{ then } \omega(p) > 0, \text{ thus subsidence.}$$

Now, considering that $\omega=0$ at the surface and at the tropopause (a good approximation at synoptic scales unless there is a strongly sloped surface), then the continuity equation tells us that the sign of divergence must change at least once in the column given these boundary conditions. This can be shown by rearranging the continuity equation:

$$\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) = - \frac{\partial \omega}{\partial p}$$

Since w must return to 0 at the boundaries, then the sign of the left side of the equation must change at least once in the equation, meaning that divergence must be 0 at some point in the profile. This level is known as a level of non-divergence (LND) since there is no divergence or convergence. At LND, there is a local maximum or minimum of ω ($\frac{\partial \omega}{\partial p} = 0$).

From synoptic experience, the LND exists in the mid-troposphere, around 500-600 hPa.



This has several implications:

- Except over sloping terrain, the divergence must change signs at least once in a column (even over sloping terrain this is common).
- There is a maximum or minimum in ω at the LND.
- Rising motion is almost always accompanied by divergence aloft and convergence below
- Subsidence is almost always accompanied by convergence aloft and divergence below.

VORTICITY

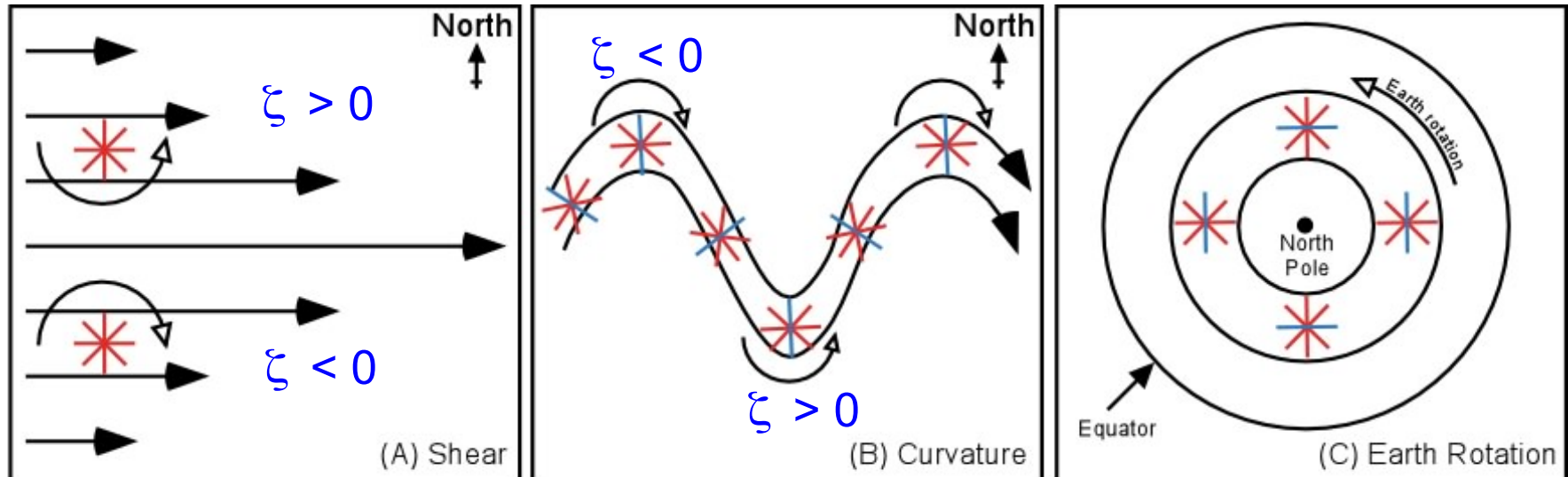
Vertical vorticity (spin about a vertical axis) arises from three sources: Horizontally sheared flow, flow curvature, and the rotation of the earth.

Relative vorticity: shear and curvature.

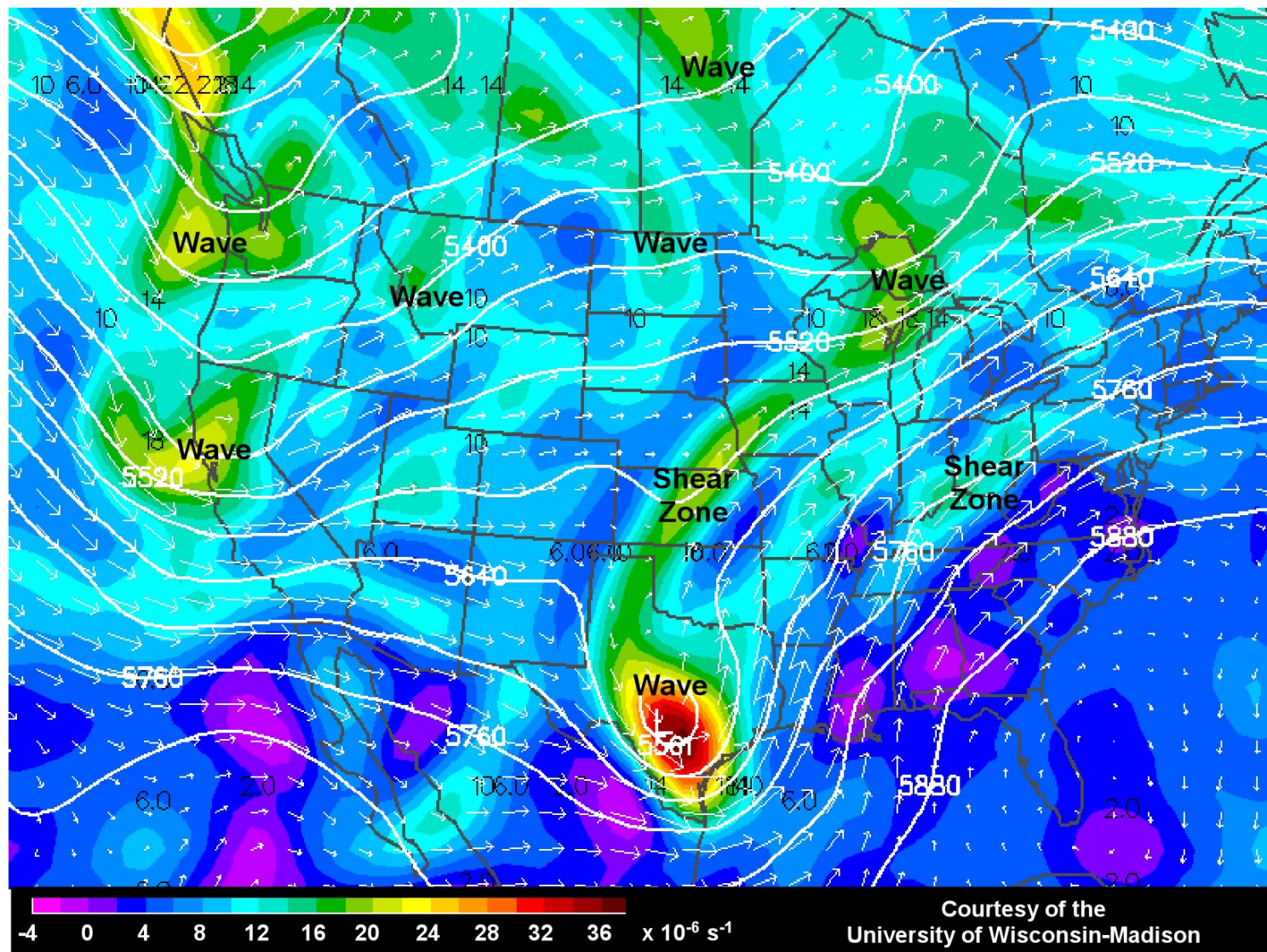
$$\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$$

Absolute vorticity: shear, curvature and earth rotation.

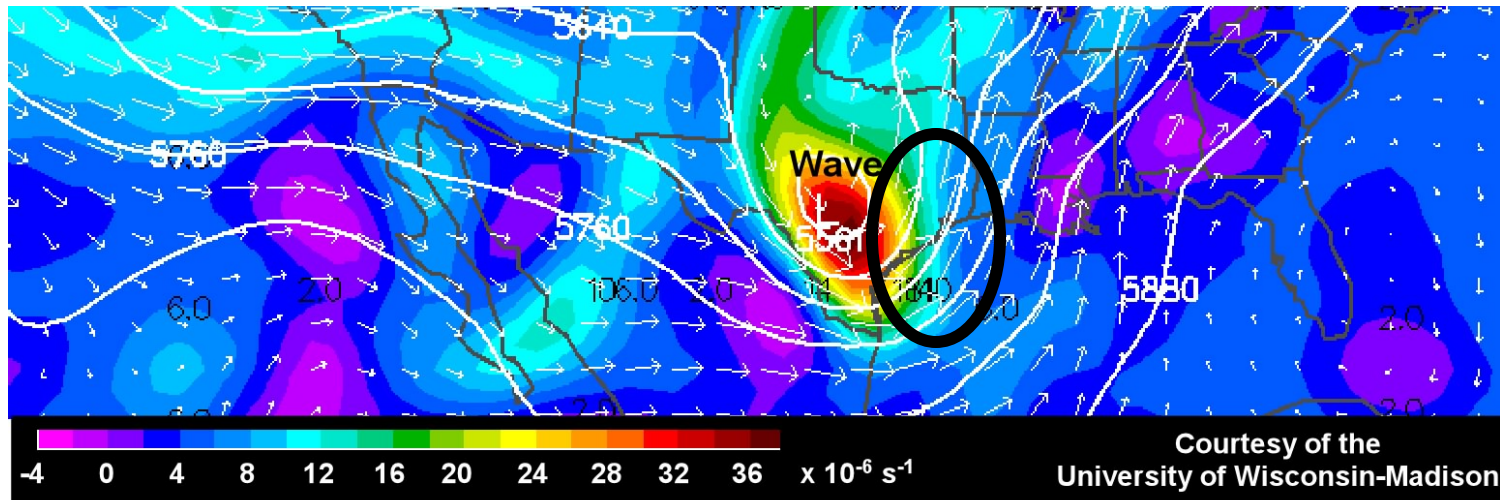
$$\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} + f$$



Absolute vorticity allows us to identify short waves and shear zones within the jetstream. Short waves trigger cyclogenesis and, in the warm season, can help trigger deep convection.



Positive Vorticity Advection on a 500 mb map can be used as a proxy for the development of low surface pressure and upward air motion.



Cartesian expression for vorticity

3D relative vorticity vector $\vec{v} = \nabla \times \vec{V} = \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) \hat{i} + \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) \hat{j} + \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \hat{k}$

Vertical component of vorticity vector (rotation in a horizontal plane)

$$\xi = \hat{k} \cdot \vec{v} = \hat{k} \cdot \nabla \times \vec{V} = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$$

Absolute vorticity (flow + earth's vorticity)

$$\eta = \hat{k} \cdot \vec{v}_a = \hat{k} \cdot \nabla \times \vec{V}_a$$

Absolute vorticity

$$\eta = \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) + f$$

The vorticity equation in height coordinates

$$\frac{du}{dt} - f v = -\frac{1}{\rho} \frac{\partial p}{\partial x} \quad (1)$$

$$\frac{dv}{dt} + f u = -\frac{1}{\rho} \frac{\partial p}{\partial y} \quad (2)$$

Expand total derivative

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} - f v = -\frac{1}{\rho} \frac{\partial p}{\partial x} + F_x \quad \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} + f u = -\frac{1}{\rho} \frac{\partial p}{\partial y} + F_y$$

Take $\frac{\partial(2)}{\partial x} - \frac{\partial(1)}{\partial y}$

write relative vorticity $\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$ as ξ

$$\frac{\partial \xi}{\partial t} + u \frac{\partial \xi}{\partial x} + v \frac{\partial \xi}{\partial y} + w \frac{\partial \xi}{\partial z} + (\xi + f) \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) + \left(\frac{\partial w}{\partial x} \frac{\partial v}{\partial z} - \frac{\partial w}{\partial y} \frac{\partial u}{\partial z} \right) + u \frac{\partial f}{\partial x} + v \frac{\partial f}{\partial y} = \frac{1}{\rho^2} \left(\frac{\partial \rho}{\partial x} \frac{\partial p}{\partial y} - \frac{\partial \rho}{\partial y} \frac{\partial p}{\partial x} \right) + \left[\frac{\partial F_x}{\partial x} - \frac{\partial F_y}{\partial y} \right]$$

$$\frac{d(\xi + f)}{dt} = -(\xi + f) \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) - \left(\frac{\partial w}{\partial x} \frac{\partial v}{\partial z} - \frac{\partial w}{\partial y} \frac{\partial u}{\partial z} \right) + \frac{1}{\rho^2} \left(\frac{\partial \rho}{\partial x} \frac{\partial p}{\partial y} - \frac{\partial \rho}{\partial y} \frac{\partial p}{\partial x} \right) + \left[\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right]$$

$$\frac{d(\xi + f)}{dt} = \underbrace{-\left(\xi + f\right)\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right)}_{\text{Divergence acting on Absolute vorticity (twirling skater effect)}} - \underbrace{\left(\frac{\partial w}{\partial x} \frac{\partial v}{\partial z} - \frac{\partial w}{\partial y} \frac{\partial u}{\partial z}\right)}_{\text{Tilting of vertically sheared flow}} + \underbrace{\frac{1}{\rho^2} \left(\frac{\partial \rho}{\partial x} \frac{\partial p}{\partial y} - \frac{\partial \rho}{\partial y} \frac{\partial p}{\partial x}\right)}_{\text{Pressure/density solenoids}} + \underbrace{\left[\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\right]}_{\text{Gradients in force Of friction}}$$

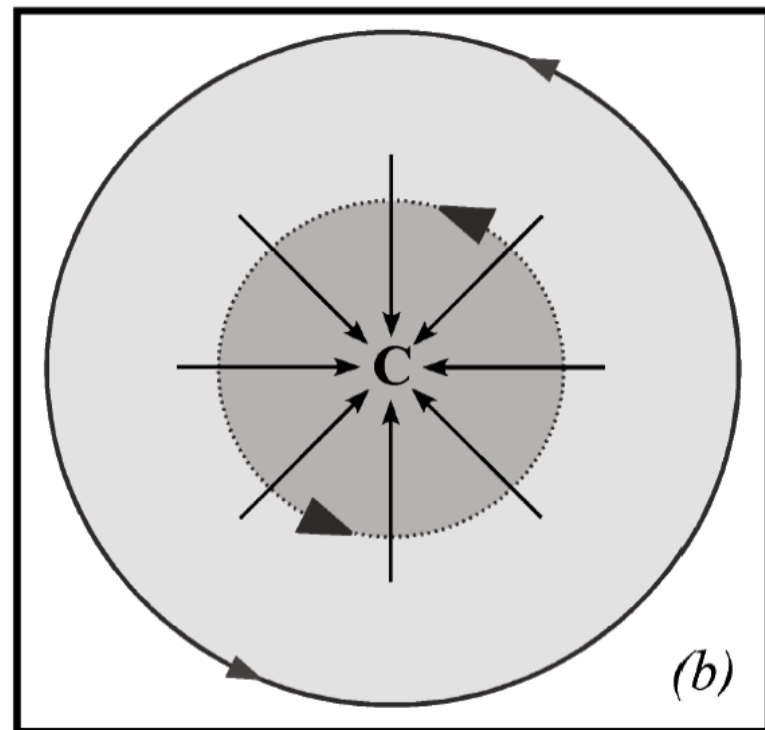
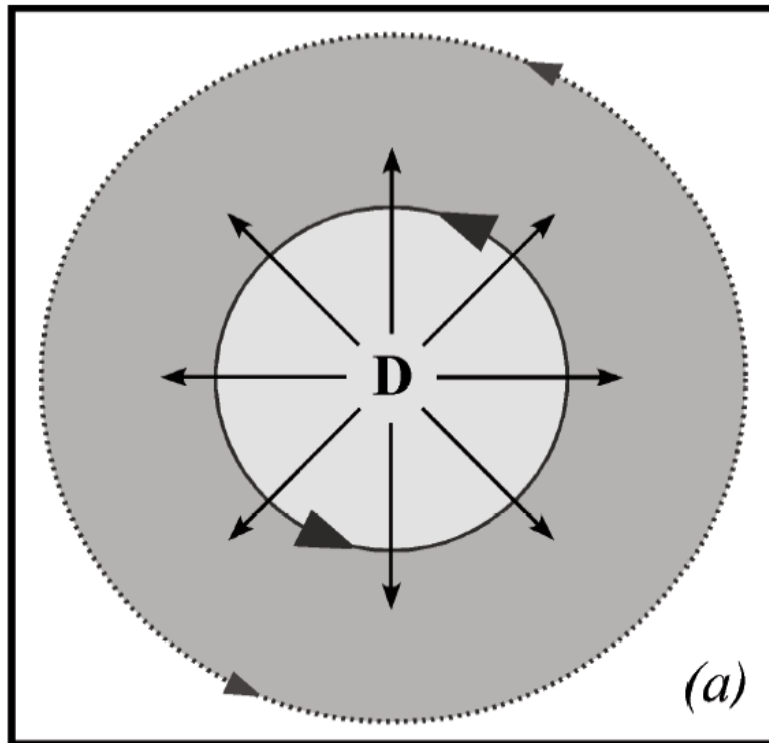
Rate of change
of relative vorticity
Following parcel

Divergence acting on
Absolute vorticity
(twirling skater effect)

Tilting of vertically
sheared flow

Pressure/density
solenoids

Gradients in force
Of friction



$$\frac{d(\xi + f)}{dt} = -(\xi + f) \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) - \left(\frac{\partial w}{\partial x} \frac{\partial v}{\partial z} - \frac{\partial w}{\partial y} \frac{\partial u}{\partial z} \right) + \frac{1}{\rho^2} \left(\frac{\partial \rho}{\partial x} \frac{\partial p}{\partial y} - \frac{\partial \rho}{\partial y} \frac{\partial p}{\partial x} \right) + \left[\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right]$$

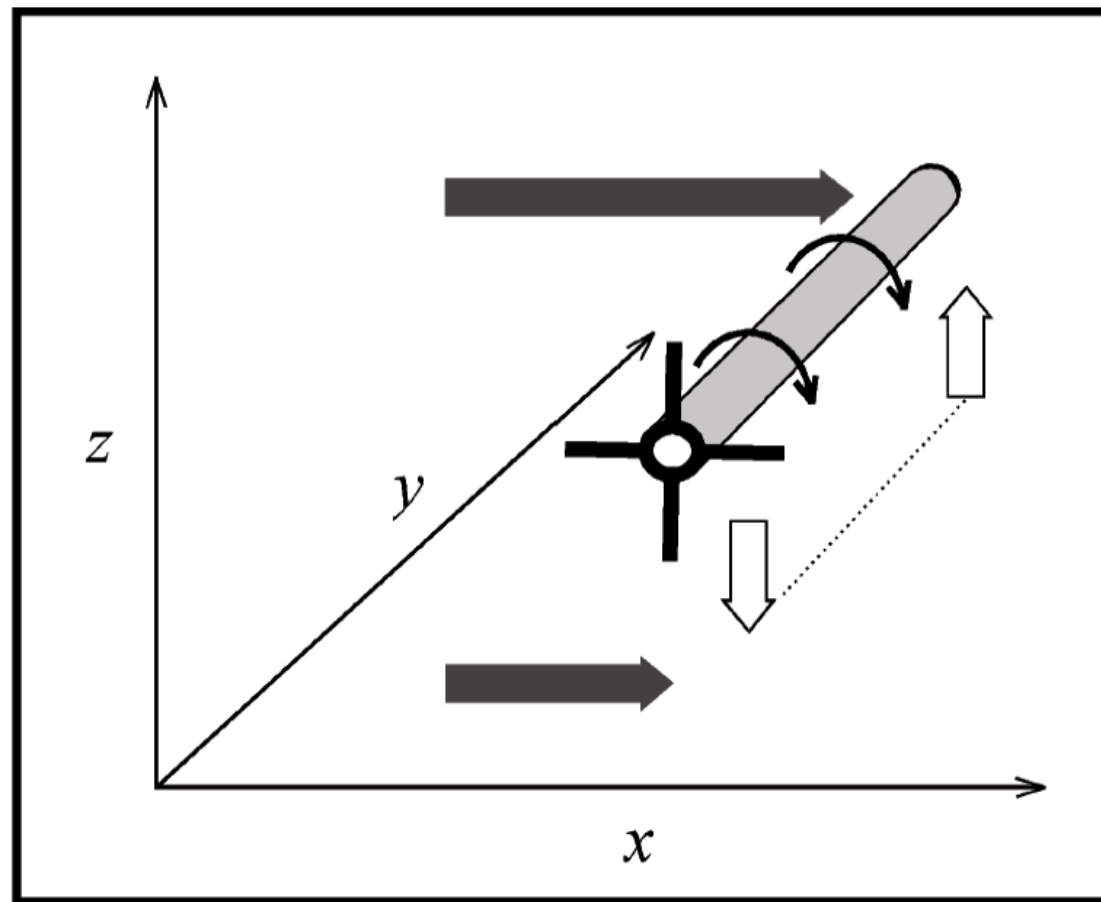
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Tilting of vertically
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Pressure/density
solenoids

Gradients in force
Of friction



$$\frac{d(\zeta + f)}{dt} = -(\zeta + f) \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) - \left(\frac{\partial w}{\partial x} \frac{\partial v}{\partial z} - \frac{\partial w}{\partial y} \frac{\partial u}{\partial z} \right) + \frac{1}{\rho^2} \left(\frac{\partial \rho}{\partial x} \frac{\partial p}{\partial y} - \frac{\partial \rho}{\partial y} \frac{\partial p}{\partial x} \right) + \left[\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right]$$

Rate of change
of relative vorticity
Following parcel

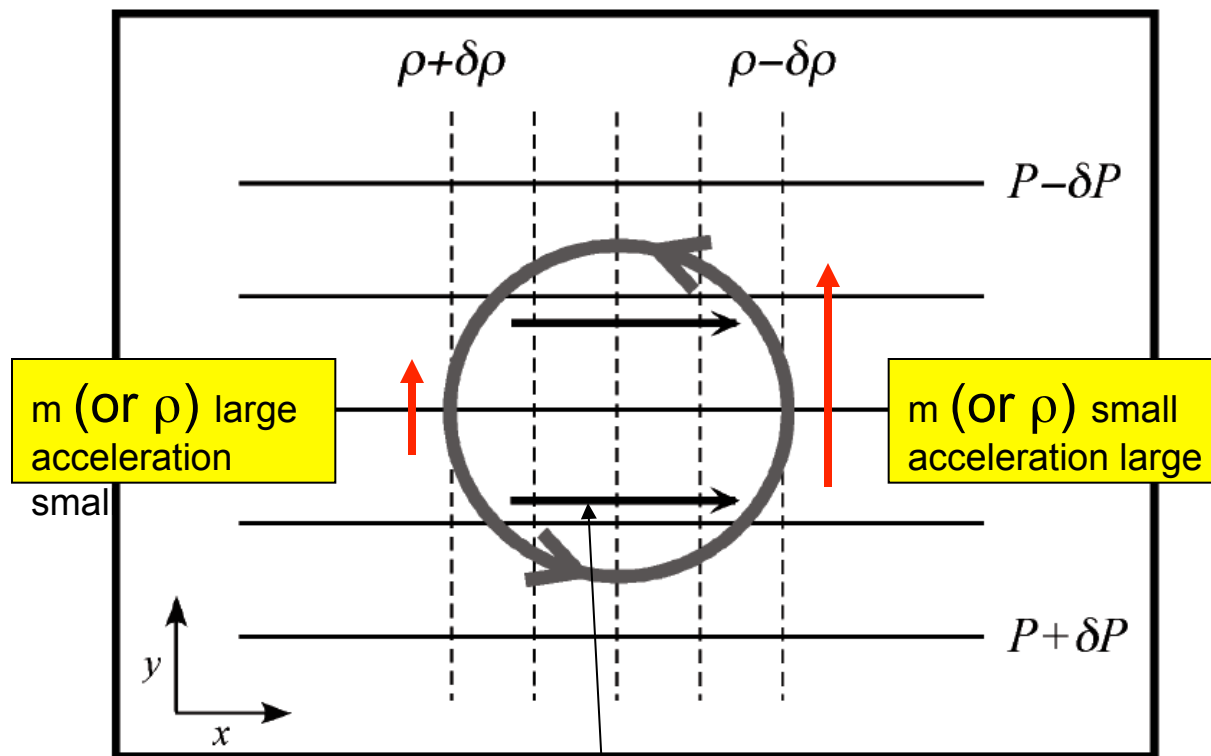
Divergence acting on
Absolute vorticity
(twirling skater effect)

Tilting of vertically
sheared flow

Pressure/density
solenoids

Gradients in force
Of friction

Solenoid: field loop that converts potential energy to kinetic energy



$$F = ma$$

$$\frac{PGF}{m} = a$$

Cold advection
pattern

geostrophic wind

$$\frac{d(\zeta + f)}{dt} = -(\zeta + f) \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) - \left(\frac{\partial w}{\partial x} \frac{\partial v}{\partial z} - \frac{\partial w}{\partial y} \frac{\partial u}{\partial z} \right) + \frac{1}{\rho^2} \left(\frac{\partial \rho}{\partial x} \frac{\partial p}{\partial y} - \frac{\partial \rho}{\partial y} \frac{\partial p}{\partial x} \right) + \left[\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right]$$

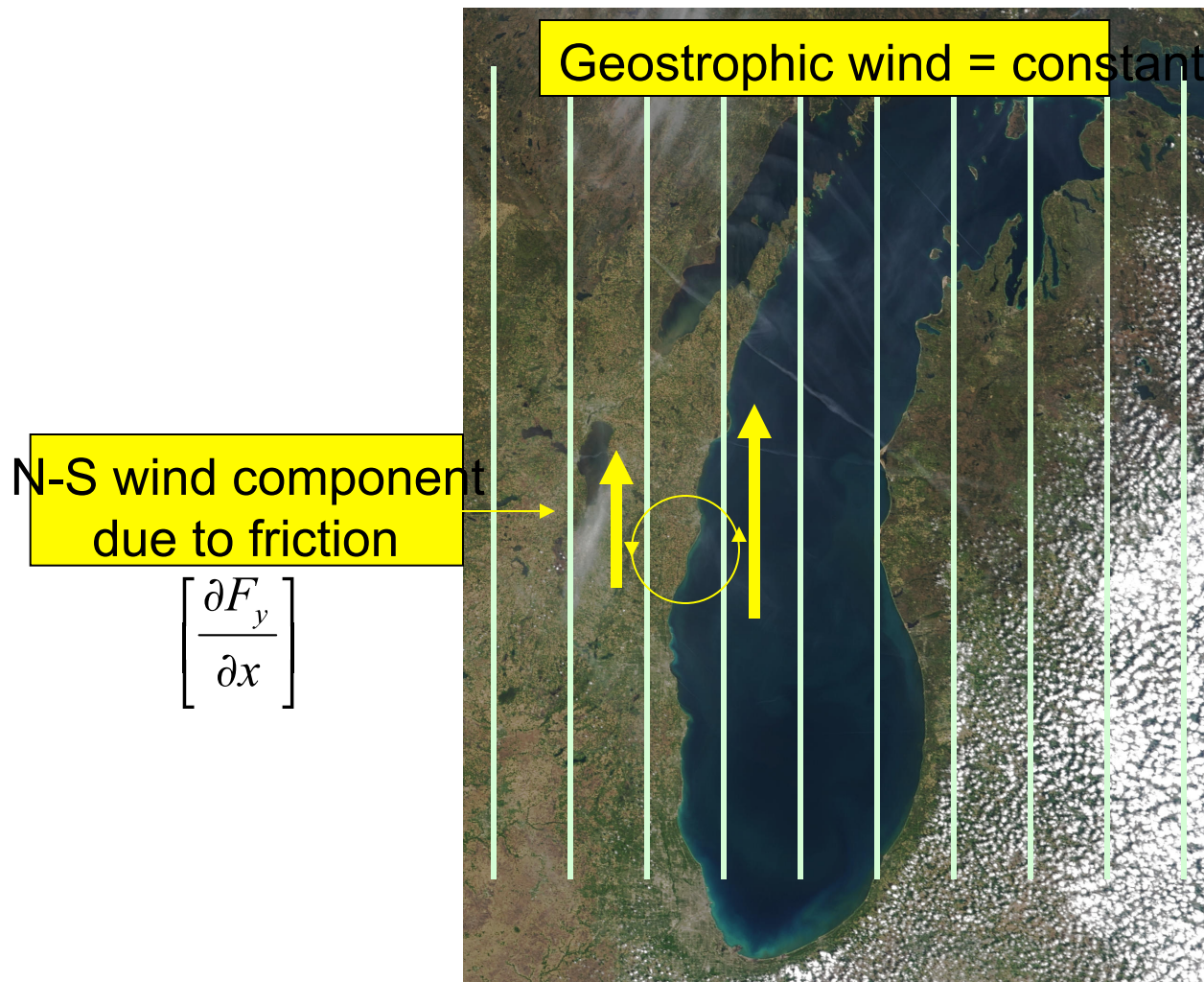
Rate of change
of relative vorticity
Following parcel

Divergence acting on
Absolute vorticity
(twirling skater effect)

Tilting of vertically
sheared flow

Pressure/density
solenoids

Gradients in force
Of friction



The vorticity equation in pressure coordinates

$$\frac{du}{dt} = -\frac{\partial\phi}{\partial x} + f v + F_x \quad (1)$$

$$\frac{dv}{dt} = -\frac{\partial\phi}{\partial y} - f u + F_y \quad (2)$$

Expand total derivative

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + \omega \frac{\partial u}{\partial P} = -\frac{\partial\phi}{\partial x} + f v + F_x \quad \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + \omega \frac{\partial v}{\partial P} = -\frac{\partial\phi}{\partial y} - f u + F_y$$

Take $\frac{\partial(2)}{\partial x} - \frac{\partial(1)}{\partial y}$

$$\begin{aligned} \frac{\partial}{\partial t} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) &= -u \frac{\partial}{\partial x} \left[\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} + f \right] - v \frac{\partial}{\partial y} \left[\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} + f \right] - \omega \frac{\partial}{\partial P} \left[\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right] \\ &\quad - \left[\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} + f \right] \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) + \left[\frac{\partial u}{\partial P} \frac{\partial \omega}{\partial y} - \frac{\partial v}{\partial P} \frac{\partial \omega}{\partial x} \right] + \left[\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right] \end{aligned}$$

write relative vorticity $\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$ as ξ

$$\frac{\partial \xi}{\partial t} = -u \frac{\partial}{\partial x} [\xi + f] - v \frac{\partial}{\partial y} [\xi + f] - \omega \frac{\partial \xi}{\partial P} - [\xi + f] \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) + \left[\frac{\partial u}{\partial P} \frac{\partial \omega}{\partial y} - \frac{\partial v}{\partial P} \frac{\partial \omega}{\partial x} \right] + \left[\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right]$$

The vorticity equation

$$\frac{\partial \zeta}{\partial t} = -u \frac{\partial}{\partial x} [\zeta + f] - v \frac{\partial}{\partial y} [\zeta + f] - \omega \frac{\partial \zeta}{\partial P} - [\zeta + f] \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) + \left[\frac{\partial u}{\partial P} \frac{\partial \omega}{\partial y} - \frac{\partial v}{\partial P} \frac{\partial \omega}{\partial x} \right] + \left[\frac{\partial F_x}{\partial x} - \frac{\partial F_y}{\partial y} \right]$$

Diagram illustrating the components of the vorticity equation:

- $\frac{\partial \zeta}{\partial t}$: Local rate of change of relative vorticity
- $-u \frac{\partial}{\partial x} [\zeta + f] - v \frac{\partial}{\partial y} [\zeta + f]$: Horizontal advection of absolute vorticity on a pressure surface
- $\omega \frac{\partial \zeta}{\partial P}$: Vertical advection of relative vorticity
- $[\zeta + f] \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right)$: Divergence acting on Absolute vorticity (twirling skater effect)
- $\left[\frac{\partial u}{\partial P} \frac{\partial \omega}{\partial y} - \frac{\partial v}{\partial P} \frac{\partial \omega}{\partial x} \right]$: Tilting of vertically sheared flow
- $\left[\frac{\partial F_x}{\partial x} - \frac{\partial F_y}{\partial y} \right]$: Gradients in force Of friction

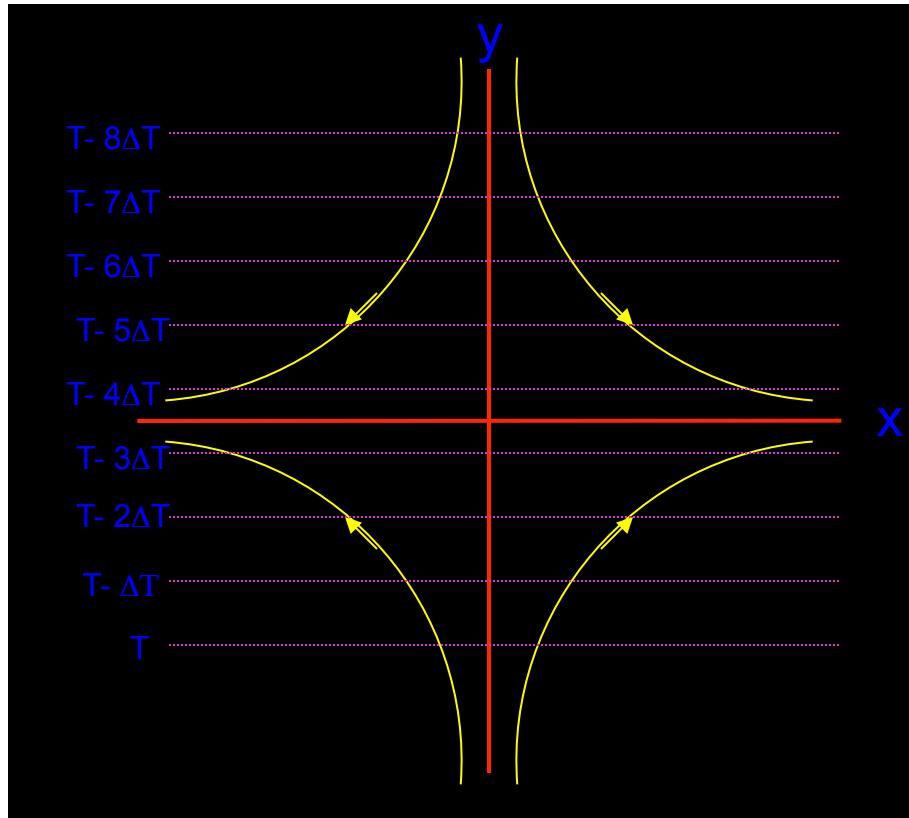
In English: Horizontal relative vorticity is increased at a point if

- 1) positive vorticity is advected to the point along the pressure surface,
- 2) or advected vertically to the point,
- 3) if air rotating about the point undergoes convergence (like a skater twirling)
- 4) if vertically sheared wind is tilted into the horizontal due a gradient in vertical velocity
- 5) if the force of friction varies in the horizontal.

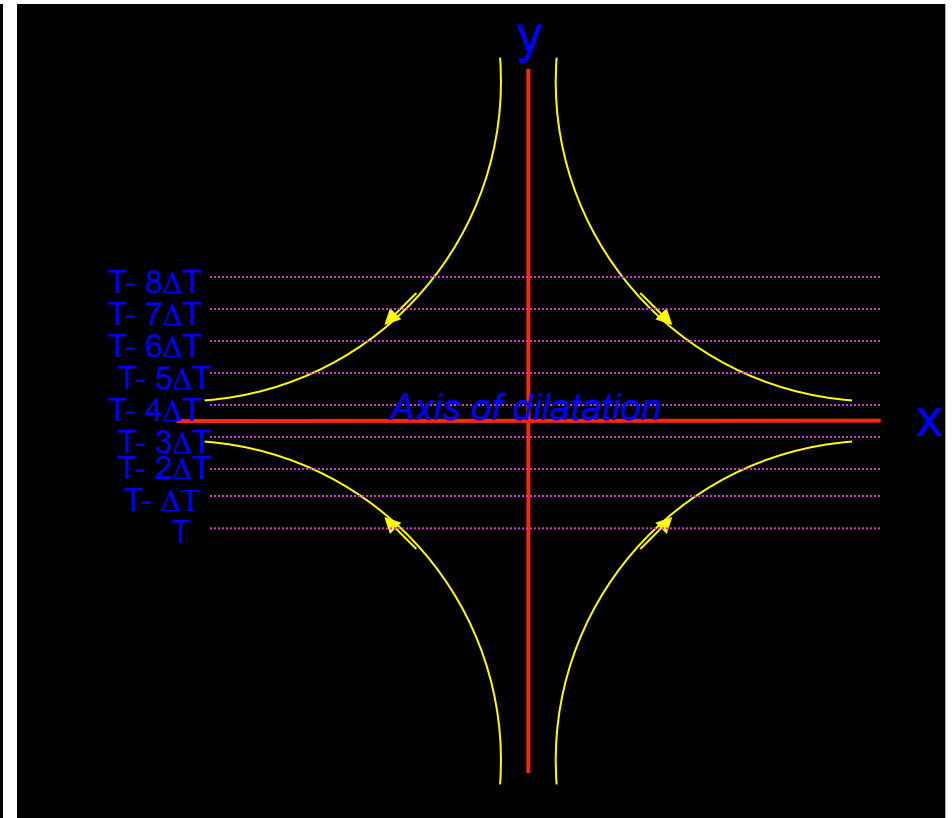
Solenoid terms disappear in pressure coordinates: we will work in P coordinate from now on

Deformation flow is fundamental to the development of fronts

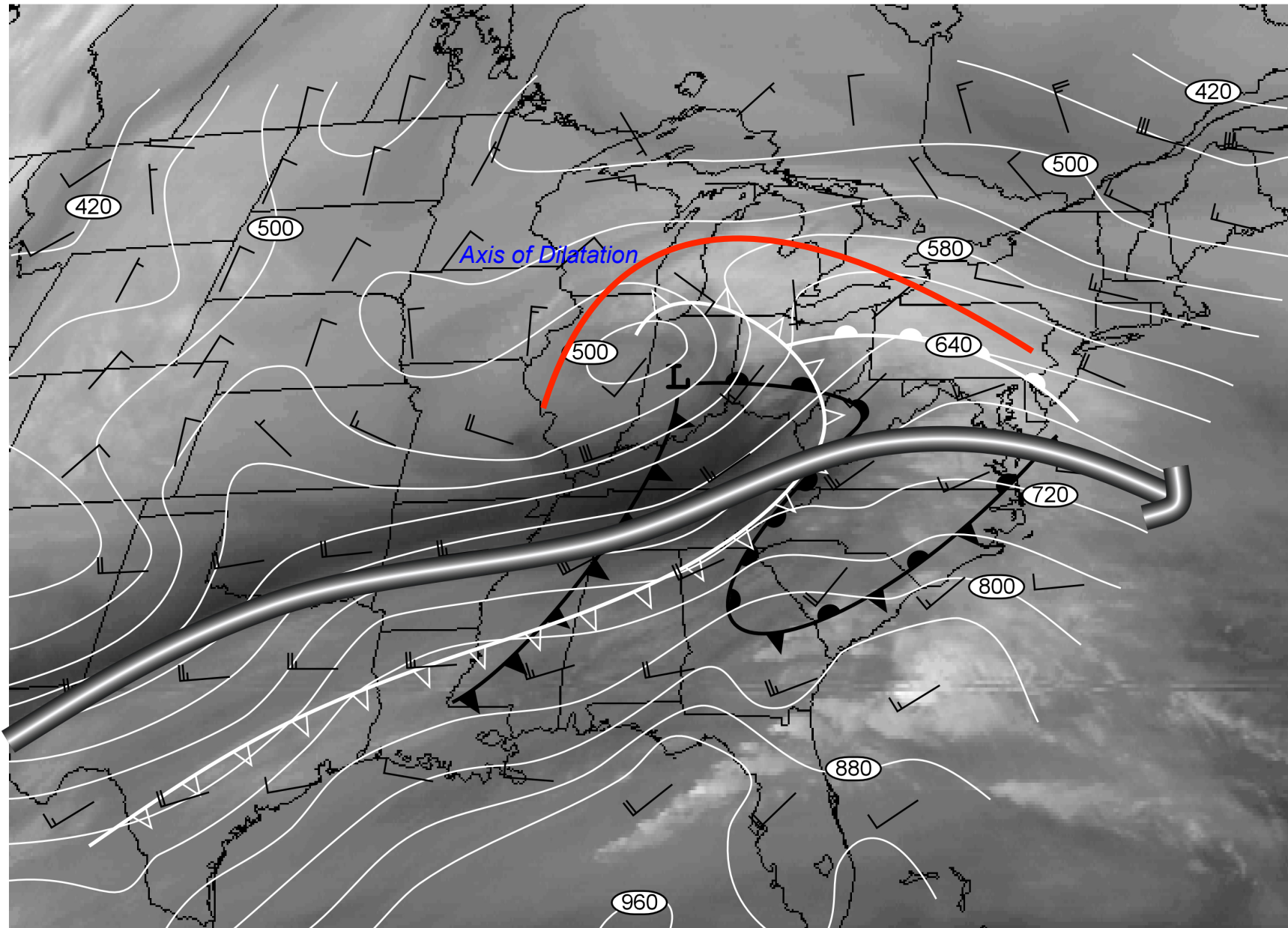
Time = t



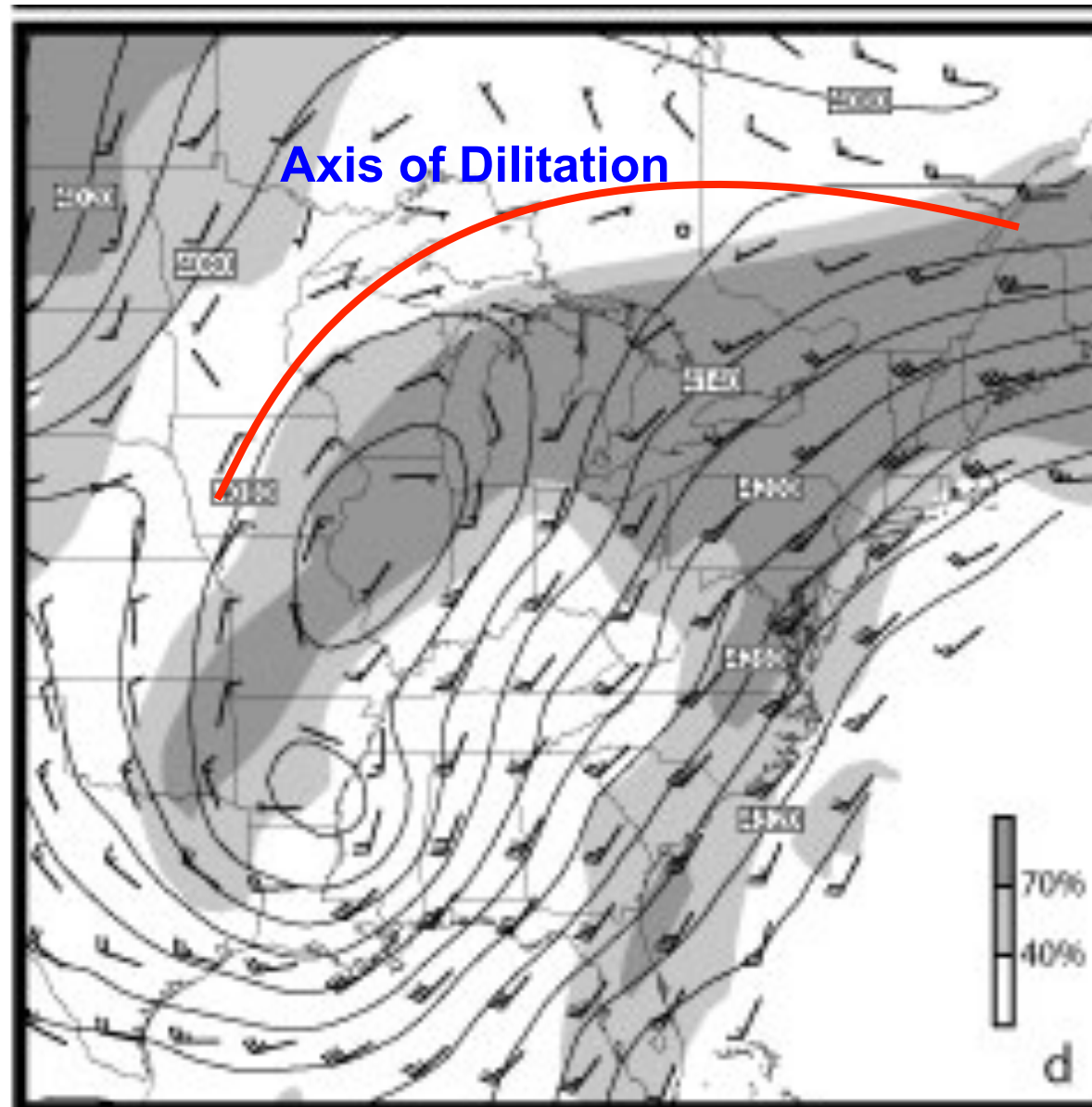
Time = $t + \Delta t$



EXAMPLES OF DEFORMATION

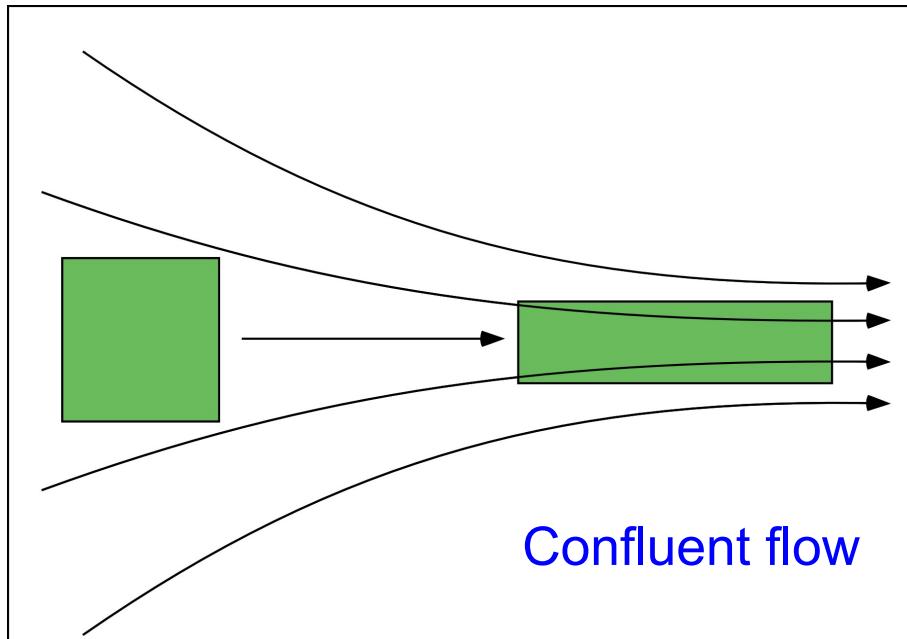


EXAMPLES OF DEFORMATION

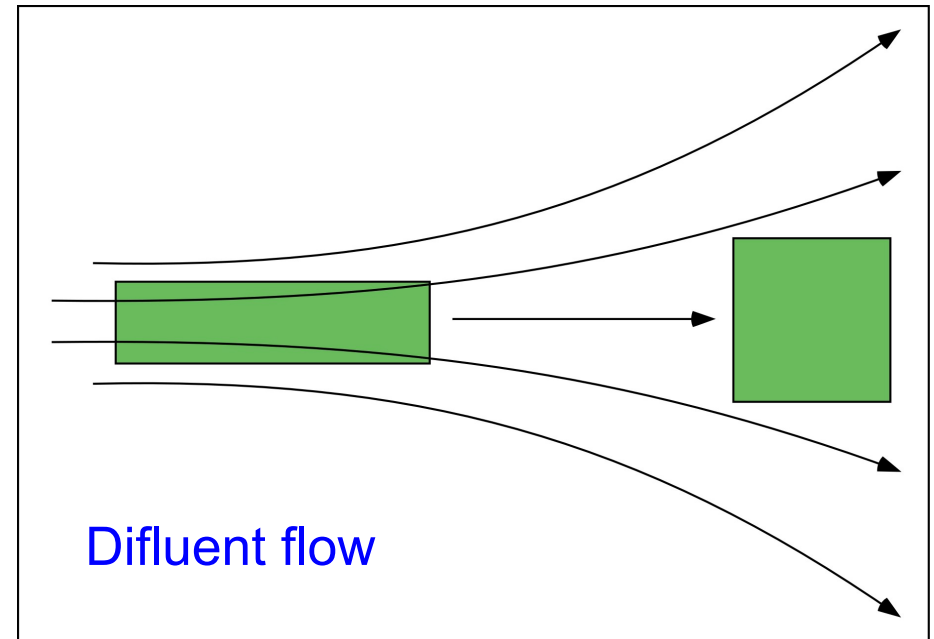


CONFLUENT and DIFLUENT FLOW

Is this flow convergent?



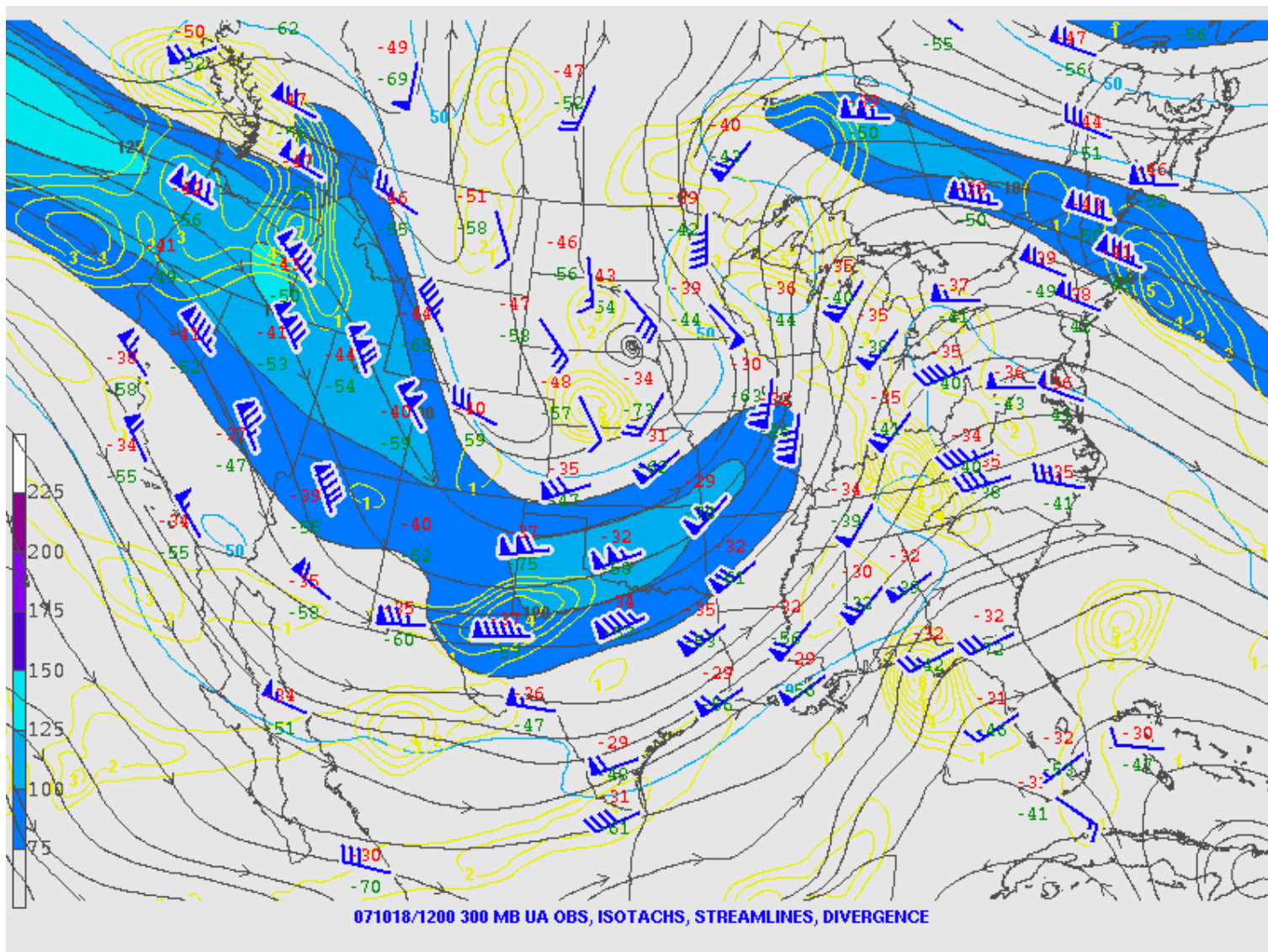
Is this flow divergent?



**NO: The areas of the two boxes are identical.
The flow is a combination of translation and
deformation.**

The terms for divergence, relative vorticity, and deformation strictly apply on a plane tangent to the earth's surface. If we take earth's curvature into account, we have to add an additional term.





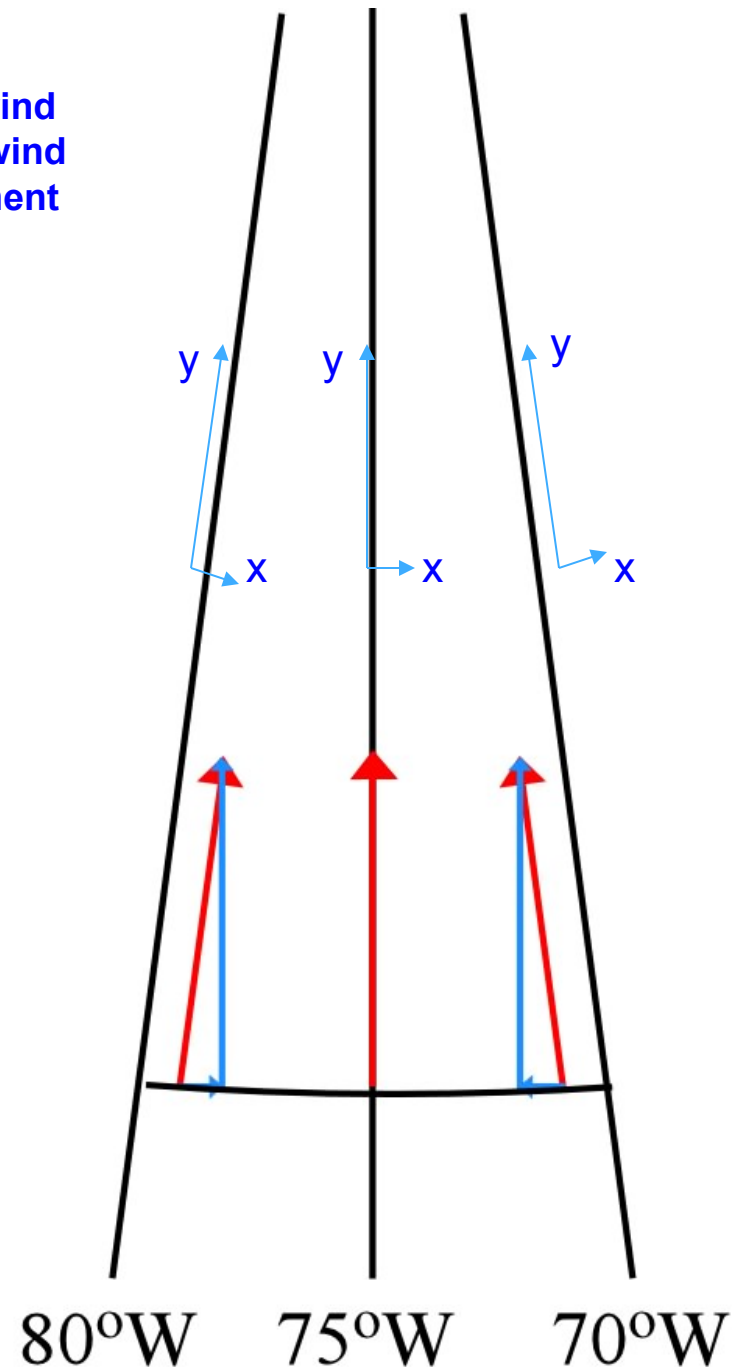
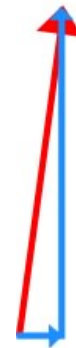
Suppose the wind is southerly and uniform. Is the wind convergent?

Yes!

$$\delta = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{v}{a} \tan \phi$$

Convergence of meridians toward north leads to convergence. This is the earth curvature term (the last term) in the expression for convergence (δ).

Red = wind
Blue = wind component

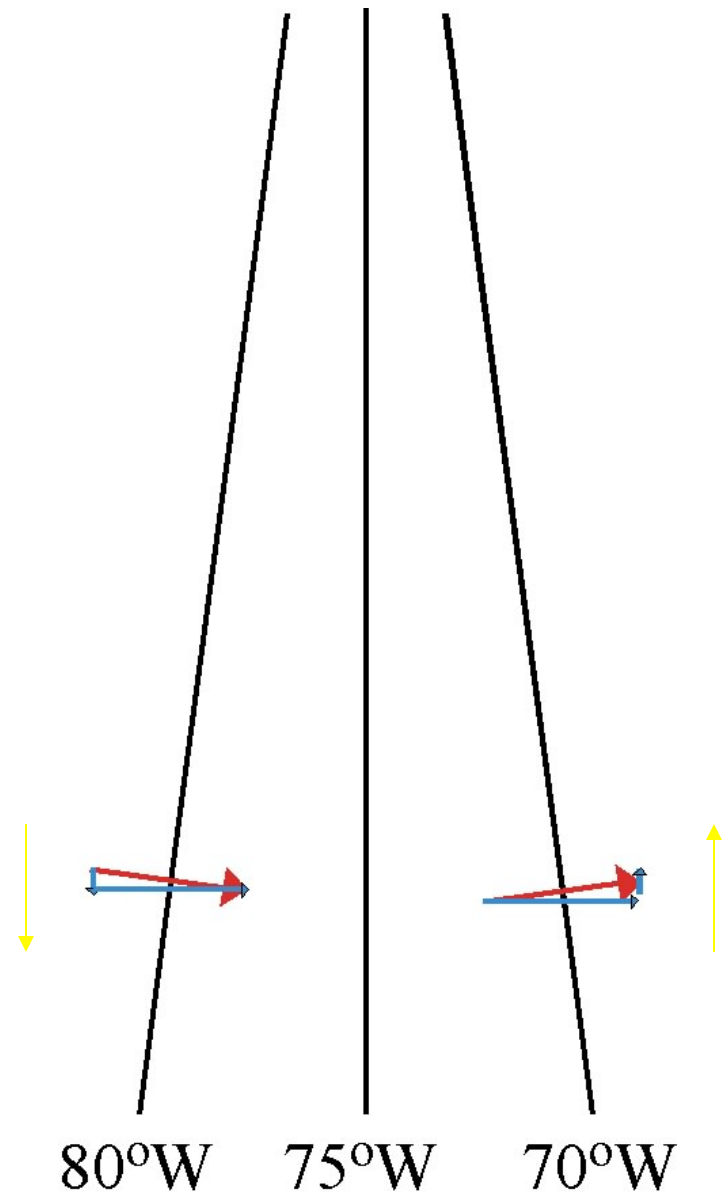


Suppose the wind is westerly and uniform. Does vorticity exist?

Yes!

$$\xi = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} + \frac{u}{a} \tan \phi$$

Convergence of meridians toward north creates vorticity. This is the earth curvature term (the last term) in the expression for vorticity (ξ).



In a similar way, convergence of the earth's meridians toward the north leads to deformation in otherwise uniform flow

$$D_1 = \frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} - \frac{v}{a} \tan \phi$$

$$D_2 = \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} + \frac{u}{a} \tan \phi$$

Earth's curvature terms are an order of magnitude smaller than other terms, but cannot be ignored in models, at least in the middle and high latitudes.